

Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-08-06 07:30:02

Situation Summary

Stage IV: 4 of 20

Biomarker Count: 14 entries

Top Targets: MSI/MMR (4) · BRAF (4) · ctDNA (2)

Breakthrough: 6 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

HIGH

All Stages

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED JUL 17, 2026

9

Circulating tumor DNA to predict the risk of venous thromboembolism in locally advanced rectal cancer

Higher baseline circulating tumor DNA (ctDNA) levels in KRAS-mutant locally advanced rectal cancer (LARC) may correlate with an increased risk of venous thromboembolism (VTE). These exploratory findings suggest a potential biomarker for VTE risk, warranting validation in larger prospective studies.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage IV

MSI/MMR

VERIFIED

RESEARCH

PUBLISHED JUL 20, 2026

8

First-Line Immunotherapy Versus Chemotherapy in MSI-H/dMMR Metastatic Colorectal Cancer: A Systematic Review and Meta-Analysis of Phase III Studies

First-line immune checkpoint inhibitor (ICI) therapy demonstrates significant survival benefits in metastatic colorectal cancer (mCRC) with microsatellite instability-high (MSI-H) or mismatch repair deficiency (dMMR), regardless of BRAF or KRAS/NRAS mutation status and tumor location. This systematic review and meta-analysis of Phase III studies supports ICI therapy as the standard of care for this patient population.

pubmed.ncbi.nlm.nih.gov

HIGH

Early Detection

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED JUL 15, 2026

7

Translational Validation of a Novel Multi-Locus ctDNA Methylation Assay for Early Detection and Stratification of Colorectal Cancer: An Exploratory Prospective, Case-Control Study

A multi-locus circulating tumor DNA (ctDNA) methylation assay demonstrated diagnostic potential for early detection and stratification of colorectal cancer in a prospective, single-center, case-control study involving 35 patients and 57 healthy controls. Blood samples were collected prior to surgical intervention or screening colonoscopy, indicating the assay's relevance in distinguishing between cancerous and non-cancerous states.

pubmed.ncbi.nlm.nih.gov

HIGH **Stage IV** **KRAS** **VERIFIED** **LIQUID BIOPSY** PUBLISHED JUL 20, 2026

5

First report of dual KRAS Y96C/Y96S resistance mutations detected by liquid biopsy in KRAS G12C-mutant metastatic colorectal cancer

Polyclonal KRAS Y96S/C mutations were identified via liquid biopsy in KRAS G12C-mutant metastatic colorectal cancer, indicating a resistance mechanism that undermines intensified vertical pathway blockade efficacy. The study highlights the discordance between tissue and liquid biopsy results, emphasizing the importance of plasma-based monitoring to address spatial heterogeneity. Next-generation inhibitors with unique binding modes may be necessary to overcome the challenges posed by Switch II pocket remodeling.

pubmed.ncbi.nlm.nih.gov

HIGH **All Stages** **KRAS** **VERIFIED** **LIQUID BIOPSY** PUBLISHED JUL 15, 2026

4

Oncogenic KRAS Promotes Extracellular DNA Clearance to Drive Immunosuppression in Colorectal Cancer

Oncogenic KRAS enhances the clearance of extracellular tumor-derived DNA (ctDNA), contributing to immunosuppression in colorectal cancer. This study highlights the dual role of ctDNA in both tumor monitoring and its potential impact on the tumor microenvironment, suggesting that KRAS mutations may influence immune evasion mechanisms. Further exploration of ctDNA's role could inform therapeutic strategies targeting immune modulation in KRAS-driven colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM **All Stages** **MSI/MMR** **VERIFIED** **RESEARCH** PUBLISHED JUL 30, 2026

3

Independent and sex-stratified association between microsatellite instability and peripheral hemoglobin in colorectal cancer

Lower peripheral hemoglobin (HB) levels are independently associated with microsatellite instability-high (MSI-H) status in colorectal cancer, though the discrimination is modest. While HB may serve as an adjunctive signal, it is not a substitute for tumor-based MMR/MSI testing. Further prospective external validation is necessary.

pubmed.ncbi.nlm.nih.gov

MEDIUM **All Stages** **HER2** **VERIFIED** **GENOMIC PROFILING** PUBLISHED JUL 27, 2026

3

HER2 overexpression/amplification in colorectal cancer: spatial intratumoral heterogeneity as a key determinant of detection discordance and therapy response

HER2 overexpression and amplification in colorectal cancer show significant spatial intratumoral heterogeneity, impacting detection concordance between immunohistochemistry (IHC)/fluorescence in situ hybridization (FISH) and next-generation sequencing (NGS). This study, utilizing two retrospective cohorts, highlights the importance of considering SIH in HER2 testing to improve therapeutic response to HER2-targeted therapies.

pubmed.ncbi.nlm.nih.gov

MEDIUM **All Stages** **BROAD SCOPE** **VERIFIED** **GENOMIC PROFILING** PUBLISHED JUL 21, 2026

3

Outcomes and Molecular Correlates of Cytoreductive Surgery and Hyperthermic Intraperitoneal Chemotherapy for Colorectal Peritoneal Metastases

BRAF V600E and KRAS G12V mutations are associated with poorer outcomes in patients undergoing cytoreductive surgery and hyperthermic intraperitoneal chemotherapy for colorectal peritoneal metastases, while KRAS G12D mutation shows a neutral prognosis. These molecular features, in conjunction with clinical and pathological factors, should inform patient selection to enhance treatment outcomes.

pubmed.ncbi.nlm.nih.gov

MEDIUM **All Stages** **BRAF** **VERIFIED** **RESEARCH** PUBLISHED JUL 20, 2026

3

Prognostic Relevance of Surgical Margins According to KRAS and BRAF Status in Colorectal Liver Metastases: BRAF-KRAS LiverMet Collaborative Group

In colorectal liver metastases (CRLMs), the prognostic significance of surgical margins varies with tumor biology. For BRAF/KRAS wild-type tumors, achieving R0 resection is optimal. However, the necessity for wider margins in high-risk molecular subgroups remains to be validated. Preoperative molecular profiling may enhance individualized surgical strategies.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

MSI/MMR

VERIFIED

GENOMIC PROFILING

PUBLISHED JUL 28, 2026

2

Unique characteristics of Lynch syndrome-associated mismatch repair-deficient colorectal intramucosal carcinoma

Lynch syndrome-associated mismatch repair-deficient (MMR-D) intramucosal carcinomas (IMCs) exhibit distinct clinicopathologic and immunophenotypic profiles compared to MMR-proficient (MMR-P) and sporadic MMR-D IMCs. These findings underscore the necessity for tailored diagnostic and management strategies for patients with Lynch syndrome.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

NGS

VERIFIED

GENOMIC PROFILING

PUBLISHED JUL 24, 2026

1

Next-Generation Sequencing of Multivergent Morphologies in Colorectal Carcinomas Identifies Unique Targetable Alterations

Colorectal carcinomas with multivergent morphology exhibit unique clinicopathologic characteristics and may harbor rare molecular drivers. This study utilized next-generation sequencing to analyze two distinct morphologic areas in 213 CRC cases, identifying potentially targetable alterations. The findings suggest that assessing nonconventional morphologic patterns can uncover actionable molecular targets in CRC.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

GENOMIC PROFILING

PUBLISHED JUL 13, 2026

1

Distinct genomic landscape of colorectal signet ring cell carcinoma reveals frequent KMT2 family alterations and SMAD4 inactivation

Colorectal signet ring cell carcinoma (SRCC) displays a unique genomic profile with frequent alterations in the KMT2 gene family and recurrent SMAD4 inactivation, indicating disruption of the TGF- β signaling pathway. The lower incidence of typical colorectal cancer driver mutations suggests alternative oncogenic mechanisms in SRCC. These insights may guide future research and therapeutic strategies tailored to this aggressive subtype.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage IV

BROAD SCOPE

VERIFIED

RESEARCH

PUBLISHED JUL 11, 2026

1

Rectal Adenocarcinoma With Enteroblastic Differentiation and Hepatic Metastases: A Rare and Aggressive Variant of Colorectal Carcinoma

Rectal adenocarcinoma with enteroblastic differentiation (CAED) can present diagnostic challenges due to the loss of typical intestinal markers and expression of oncofetal proteins. In a case of metastatic rectal CAED, actionable ERBB2 (HER2) amplification was identified, highlighting the importance of comprehensive biomarker testing in atypical presentations. This finding suggests potential therapeutic targets in a variant of colorectal cancer that is aggressive and may mimic other malignancies.

pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

ctDNA

Dynamic Circulating Tumor DNA Methylation Monitoring Guiding Postoperative Surveillance in Nonmetastatic Colorectal Cancer: A Prospective, Randomized, Phase III FIND Trial

Dynamic monitoring of circulating tumor DNA (ctDNA) methylation was assessed in a prospective, randomized, Phase III FIND trial for postoperative surveillance in nonmetastatic colorectal cancer. The study indicates that ctDNA methylation can guide surveillance strategies, potentially improving early detection of recurrence.

MSI/MMR

EMPIRE (NSABP FC-13): a biomarker-driven phase II platform trial evaluating cemiplimab-based immunotherapy in microsatellite-stable colorectal cancer with ctDNA-defined minimal residual disease

In the EMPIRE (NSABP FC-13) phase II trial, cemiplimab-based immunotherapy was evaluated in microsatellite-stable colorectal cancer patients with ctDNA-defined minimal residual disease. The study aims to identify high-risk patients post-curative therapy who may benefit from additional treatment, addressing the limited efficacy of immune checkpoint inhibitors in this population.

KRAS

Circulating tumor DNA (ctDNA) clearance as an early indicator of sotorasib + panitumumab efficacy and prognosis in KRAS G12C-mutated colorectal cancer (mCRC): Results from phase 3 CodeBreaK 300.

In the phase 3 CodeBreaK 300 trial, ctDNA clearance was assessed as an early indicator of treatment efficacy and prognosis in KRAS G12C-mutated mCRC patients receiving sotorasib and panitumumab. The findings suggest that ctDNA clearance may serve as a valuable biomarker for monitoring therapeutic response and predicting outcomes in this patient population.

TP53

Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF

Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

NGS

Rapid On-Site Next-Generation Sequencing: An Alternative to Single-Gene and Send-Out Testing in Non-Small Cell Lung Cancer and Colorectal Cancer in a Community Pathology Laboratory Setting

Next-generation sequencing (NGS) using the ThermoFisher Oncomine Precision Assay (OPA) was assessed as an alternative to single-gene panels and send-out NGS for identifying actionable mutations in colorectal cancer. The study focused on turnaround time, quantity not sufficient rates, and detection of NCCN-recommended alterations, indicating that OPA may enhance efficiency in mutation identification for targeted therapies in a community pathology laboratory setting.

RAS

The temporal evaluation of RAS mutation by liquid biopsy at progression after bevacizumab-based treatment in patients with metastatic colorectal cancer

Dynamic changes in plasma RAS mutation status during disease progression in metastatic colorectal cancer (mCRC) were observed, indicating tumor clonal evolution. Notably, RAS mutation clearance correlated with a survival advantage, suggesting prognostic relevance. Reassessing RAS status via liquid biopsy at progression may yield clinically relevant insights, though findings are exploratory and warrant further investigation.

HER2

Comprehensive genomic landscape of ERBB2 in Chinese GI tumors: mutation-centered landscapes and precision treatment opportunities

ERBB2 alterations in colorectal cancer (CRC) and gastric cancer (GC) predominantly occur in kinase domain hotspots, influenced by tumor-specific genomic contexts. Oncogenic ERBB2 mutations correlate with higher tumor mutational burden (TMB) and microsatellite instability-high (MSI-H) status. These findings support the clinical evaluation of mutation-selective HER2 inhibitors and their potential combination with immune checkpoint blockade for targeted therapy in CRC and GC.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

EGFR

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

Butyrate may serve as a predictive biomarker for treatment response in KRAS wild-type colorectal cancer, as it reduces cetuximab efficacy via EGFR and AMPK-Wip1 signaling pathways.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

TMB

Plasma-TMB (pTMB) and circulating tumor fraction (ctF) dynamics as biomarkers of benefit from the addition of atezolizumab to first-line FOLFOXIRI/bevacizumab in metastatic colorectal cancer (mCRC): A translational analysis of the AtezoTRIBE study.

In the AtezoTRIBE study, plasma tumor mutational burden (pTMB) and circulating tumor fraction (ctF) were analyzed as biomarkers for the efficacy of adding atezolizumab to first-line FOLFOXIRI/bevacizumab in metastatic colorectal cancer. The study suggests that pTMB and ctF dynamics may predict treatment benefit, providing insights for personalized therapy strategies in mCRC.

Stool DNA

Enhancing non-invasive colorectal cancer screening with stool DNA methylation markers and light gradient-boosting machine learning

Stool DNA methylation markers CNRIP1, SFRP2, and VIM demonstrated high sensitivity for non-invasive colorectal cancer screening. The application of the LightGBM machine learning algorithm improved diagnostic accuracy, suggesting a viable method for early CRC detection.

Methylation

Methylation biomarkers for early detection of colorectal cancer: From molecular discovery to clinical translation and application

Key methylation biomarkers (SEPT9, BMP3, NDRG4) have been identified for early detection of colorectal cancer (CRC) through genome-wide association studies and candidate gene approaches. These biomarkers have been validated using high-throughput technologies, facilitating their transition from research to clinical application. Early screening utilizing these methylation markers may improve patient outcomes in CRC management.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right now are **ctDNA, MSI/MMR, KRAS, TP53, and BRAF**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

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About This Report

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Limitations

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https://petroai1492.github.io/delta/timelines/coloncancer_biomarker_timeline.pdf